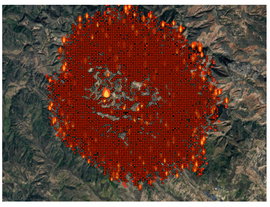
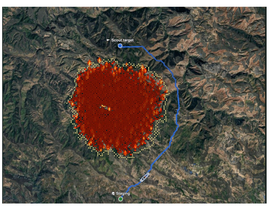
How the dog decides



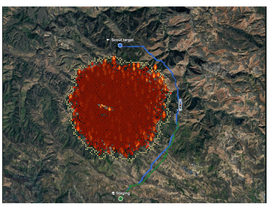
1. Predict

Forecast when each 50 m cell will burn, in 15-minute steps.



2. Plan a safe route

Pass every spot 30+ min before the fire. Brush (0.5×) and forest (0.7×) are slower; uphill is slowest. No water or slopes over 35°.



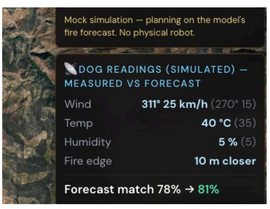
3. Keep an escape path

Aim for a way home that stays clear for 60 min, re-planned as the dog moves.



4. Scout

Walk to the fire edge and measure wind, temperature, humidity and where the edge really is.



5. Feed back

The readings correct the forecast, and the loop starts again.

Simulator views on real terrain. Robot image: design concept; sensors are simulated.

How we tested it

Two fires run at once on the same terrain:

Forecast fire
What the dog plans on.

True fire (hidden)
Hotter, drier, with a wind shift. The dog only learns about it through its readings.

Mock simulation; no physical robot yet.

Status and next steps

- ✓ ML fire forecast trained and tested
- ✓ 3D browser simulator on real terrain
- ✓ Robot-dog loop in mock simulation
- ○ Forecast + planner on the dog's controller
- ○ Real sensors feeding the forecast
- ○ Tests on real robot dogs

References

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections); Open-Meteo weather archive; ESA WorldCover 10 m land cover; Cesium World Terrain; Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.

Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*